

Research Paper

From pixels to points: An AI framework with weaker-and-fewer-labels for lightweight 3D phenotyping using 2D-3D coordinate mapping and VLMs

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ABSTRACT

3D phenotyping of seedlings is crucial to tomato cultivation in greenhouse facilities. Current studies focus on high-quality point cloud reconstruction and artificial intelligence (AI) 3D segmentation to derive phenotypic traits like plant height and crown width, which heavily rely on manual annotation and possess high complexity in deployment. This study proposes a novel AI framework from pixels to points, for efficient 3D plant phenotyping of tomato seedlings. Through the integration of 2D-3D coordinate mapping and AI vision language models, the proposed method enables accurate reconstruction and analysis of 3D phenotypic traits from single-view data. Top-down RGB images and corresponding point clouds with spatial alignment are captured using a binocular camera. Vision language models are employed with the text prompt “plant” to automatically generate bounding boxes and masks, thereby minimizing manual annotation. These outputs are further transferred to a lightweight YOLO11-segment model. The core innovation is established in our 2D-3D mapping strategy, through which plant-specific 3D points are efficiently extracted using only 2D masks. Non-plant points within initial masks are repurposed to determine ground height for improved plant height estimation, while masks are refined using the Excess Green Index to enhance crown width measurement. An mAP₅₀ of 96.0% is achieved by the YOLO11-segment model. Concerning sparse canopy, highly accurate results are yielded by our phenotyping approach, with RMSE values of 1.7 cm for plant height and 1.0 cm for crown width, and R² values of 0.93 and 0.95 against manual measurements. For dense canopy, the usage of a reference chessboard improves the performance (RMSE was reduced from 9.57 cm to 2.07 cm). Annotation dependency is significantly reduced, computational complexity is decreased, edge deployment is supported, and efficient technology transfer is enabled by the presented method. Considerable potential is offered for high-throughput screening of elite tomato varieties with desirable agronomic traits.

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1. Introduction

Tomato stands as one of the three major globally traded vegetables, holding significant importance in international vegetable commerce (Su et al., 2025). Recognized as both a functional vegetable and a high-value crop in protected agriculture systems, it has gained prominence due to its rich content of bioactive compounds, including lycopene, ascorbic

acid, and flavonoids (Wu et al., 2023). These phytochemical constituents establish tomato as a nutritional powerhouse, contributing to its dual status as a dietary essential and an economically valuable agricultural commodity (Hao et al., 2025). The cultivation of tomato during the seedling stage critically determines its subsequent yield (Kasimatis et al., 2025) and plant stress resistance (Kanojia et al., 2024), making seedling selection a crucial factor for successful tomato production (Li et al., 2024). Conventional seedling screening methods relying on manual observation of phenotypic parameters like plant height, stem diameter, and leaf area suffer from inefficiency, subjectivity, and scalability

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limitations (Liang et al., 2025; Shmaryahu et al., 2024). With the transformation from traditional agriculture to smart farming systems (Das et al., 2025), there emerges an urgent need to develop rapid, accurate, and non-destructive techniques for capturing morphological and physiological information of seedlings, which is a key challenge in optimizing the seedling selection process (Bian et al., 2022; Zhang et al., 2023a).

Currently, intelligent plant phenotyping, an integration of advanced digital sensors, deep learning, and other new technologies, has become a new powerful tool for plant growth monitoring and analysis (Shi et al., 2019; Zhou et al., 2024). Among emerging techniques, the combination of RGB imaging and two-dimensional Measuring the cucumber leaf area using improved Mask R-CNN convolutional neural network (2D-CNN) has dominated plant phenotyping research due to its accessibility, low cost, and rapid processing capabilities (Vayssade et al., 2022). For instance, a YOLO-based (You Only Look Once) image processing model achieved real-time detection and segmentation of tomatoes at different mature stages, realizing remarkable mean Average Precision at 50% IoU (mAP_{50}) and F1 scores of 91.4% and 85.4%, respectively (Wang et al., 2024a). These models could also extract other plant organs, including stems, leaves, flowers, and fruits across different growth phases with satisfactory precision (Yang et al., 2024). There are also some existing visual models that produce fine-grained segmentation masks for plant phenotyping. They are commonly designed by modifying popular visual foundation models, realizing high performance on both public plant phenotyping datasets, like CVPPP Leaf Segmentation Challenge datasets (Ward and Moghadam, 2020), as well as private rice segmentation dataset (Xu et al., 2025). These high-quality segmentation masks could ensure the accuracy of subsequent phenotypic parameter calculations.

However, two-dimensional (2D) imaging can only present plant characteristics from limited perspectives, failing to capture comprehensive whole-plant information (Luo et al., 2023). Moreover, its reliance on reference objects and pixel-based measurements lacks the capability for direct physical dimension acquisition (Zhang et al., 2023b). These limitations significantly constrain the evaluation of critical agronomic traits. To this end, researchers have turned to three-dimensional (3D) sensing technologies. For instance, individual maize plant organ segmentation can be achieved by integrating Light Detection And Ranging (LiDAR) data to obtain local plant density and three-dimensional morphological structure, with accuracy in both component segmentation ($F\text{-score} = 0.82$) and plant segmentation ($F\text{-score} = 0.99$) (Ao et al., 2022). Another case is an incremental fusion method that reconstructs 3D point clouds using multi-view 2D image segmentation results, which achieved precision, recall, and F1 scores of 0.98, 0.91, and 0.94, respectively (Wang and Zhang, 2025). Additionally, a 3D point cloud convolutional neural network framework, implementing a rapid and efficient segmentation method, was innovated to tackle the difficulty of spike-shaped point cloud segmentation using traditional machine learning approaches, and the method outperforms classical PointNet models in both accuracy and Intersection over Union (IoU) metrics (Gong et al., 2021). Besides, RGB-D imaging also provides a possible solution for plant 3D analysis, which offers inherent 2D-to-3D alignment and supports multimodal learning. Intel D435i was widely used for vision-based phenotyping (Wang et al., 2024b) and fruit harvesting (Yan and Li, 2024). However, the limited depth resolution limited the performance of 3D reconstruction.

While LiDAR and multi-view imaging techniques enable 3D point cloud reconstruction (Schönberger and Frahm, 2016), their implementation faces challenges. LiDAR systems involve relatively high equipment costs, and multi-view imaging technologies require complex and time-consuming data acquisition processes. The noise-intensive nature of raw point cloud data also escalates computational complexity in processing workflows. More critically, direct analysis of such 3D data typically relies on strongly-supervised learning, which demands extensive manual annotation efforts (e.g., assigning class and instance IDs to

each 3D point). This “annotation bottleneck” severely hinders the scalability and practical deployment of 3D phenotyping. The current research mainly focuses on the following aspects. In 2D-based plant phenotyping studies, investigators prioritize model architecture optimization through mechanisms such as attention modules and specialized functional components to enhance the modeling performance (Deng et al., 2023), yet often overlook strategies to substantially reduce annotation costs (Cheng et al., 2024). As for 3D-based plant phenotyping studies, focus remains on direct segmentation of complex point clouds using sophisticated 3D deep learning networks, with limited exploration of lightweight or annotation-efficient alternatives (Zhang et al., 2024). Consequently, two urgent challenges persist in plant phenotyping research, including mitigating reliance on costly manual annotations and achieving faster 3D point cloud segmentation on edge devices.

To address the above-mentioned challenges, this study proposes a novel AI framework that significantly reduces dependency on both the quantity and complexity of annotations—i.e., it operates with weaker and fewer labels. The core of our method integrates 2D–3D coordinate mapping with vision language models (VLM). We adopted binocular vision to capture top-down RGB images and spatially aligned 3D point clouds. Instead of requiring dense 3D point labels, our approach leverages the zero-shot capability of VLMs. A simple text prompt (“plant”) was used with Grounding DINO to automatically generate initial 2D bounding boxes, which were then refined into segmentation masks by the Segment Anything Model (SAM). This process requires only minimal manual correction, representing a drastic reduction in annotation effort. These AI-generated 2D masks serve as ‘weak labels’ and are then transferred to a lightweight YOLO11-segment model for efficient inference. Finally, through a precise 2D–3D coordinate mapping strategy, the 2D masks are used to efficiently segment the corresponding 3D plant points, bypassing the need for complex 3D neural networks. This pipeline successfully achieves accurate extraction of key phenotypic parameters, such as plant height and crown width, demonstrating that highly accurate 3D phenotyping can be realized with dramatically weaker and far fewer manually annotated labels. The main contributions of this study are summarized as follows.

- (1) **Real-time low-cost 3D phenotyping of tomato plants.** Single top-view RGB imaging and point cloud reconstruction are adopted, with the advantages of high efficiency and deployment friendliness. It shows superiorities in conventional LiDAR or multi-view imaging-based 3D phenotyping, concerning their limitations, including complex equipment movement and control, complex point fusion, and high-cost laser scanning operations and devices.
- (2) **Weaker labels simplify the 3D plant segmentation.** Cutting the 3D point cloud using 2D pixel-masks (‘weaker labels’) is proposed, avoiding complex and computationally intensive 3D neural networks construction and 3D annotations (‘strong labels’), which simplifies the 3D segmentation to a 2D segmentation task.
- (3) **VLMs enable the implementation of fewer label-based modeling, and knowledge transfer further simplify the AI application.** VLMs with a huge number of parameters could generate segmentation masks for tomato plants. Only the effective knowledge is transferred to a YOLO model (famous for its tiny volume), realizing AI modeling with both low labor cost and high inference efficiency.

2. Materials and methods

2.1. Overview of the proposed method

In this study, a novel tomato seedling 3D phenotyping method was proposed for calculating the plant height and crown width parameters. Rapid 3D point cloud segmentation of the seedling was achieved using

simple 2D CNN segmentation and 2D–3D coordinate mapping. VLM was employed for generating training labels, which makes the whole method requires weaker and fewer labels, significantly reducing manual annotation work. The segmented point cloud data of a single plant was used for plant height and crown width calculation. The diagram is illustrated in Fig. 1.

Firstly, tomato plants were captured using a binocular camera to acquire 2D images and the corresponding 3D point clouds. Subsequently, a tomato plant extraction model was established using deep learning instance segmentation. Then, the 2D segmentation masks were aligned with their corresponding 3D point clouds through coordinate mapping, achieving rapid 3D point cloud segmentation driven by 2D data, avoiding the usage of complex AI models for direct 3D point cloud processing. Finally, phenotypic parameters of tomato plants were efficiently extracted according to the segmented point clouds. Detailed implementation methods are presented in Sections 2.3 and 2.4.

2.2. Image acquisition and dataset preparation

The tomato seeding plants cultivated in different regions (including one tomato cultivation greenhouse and one outdoor experimental site in Zhuji, Zhejiang Province, as well as another indoor experimental site in Nanjing Forestry University, Nanjing, Jiangsu Province) were selected for the experiment. Different varieties (Zhefen20172, Zhefen202, Jingfan603, Jinfan502) were selected to ensure sample

diversity, planted with different row spacing and plant spacing. For further evaluating the possibility of expanding the designed method for broader applications of analyzing other plant types, an extra dataset was collected including another four tomato varieties (ID-832, ID-89, ID-808, ID-815), as well as lemon trees, eggplant seedlings and passion fruit seedlings.

There are several typical 3D sensing devices for plant analysis, which are listed and compared in Table 1. Considering the cost, data collection speed, feasibility of deployment on edge computing devices, control interfaces and other key factors, ZED mini2 stereoscopic camera was selected for plant image and 3D point cloud data collection. The resolution of image acquisition was set to 1920×1080 . The imaging system utilized the manufacturer-provided SDK from Stereolab, NVIDIA CUDA acceleration, and an edge computing device, NVIDIA Jetson Orin NX (16GB RAM). Shooting in top-down mode was selected to get tomato plant information.

2.3. Segmentation of tomato plant images

2.3.1. AI models

Grounding DINO (GD) was a Transformer-based detector that could identify specified objects through human language inputs by combining the DINO self-supervised learning model with grounded pre-training (Liu et al., 2024). It adopted a “dual-encoder and single-decoder” architecture. The text backbone employed the BERT model to interpret

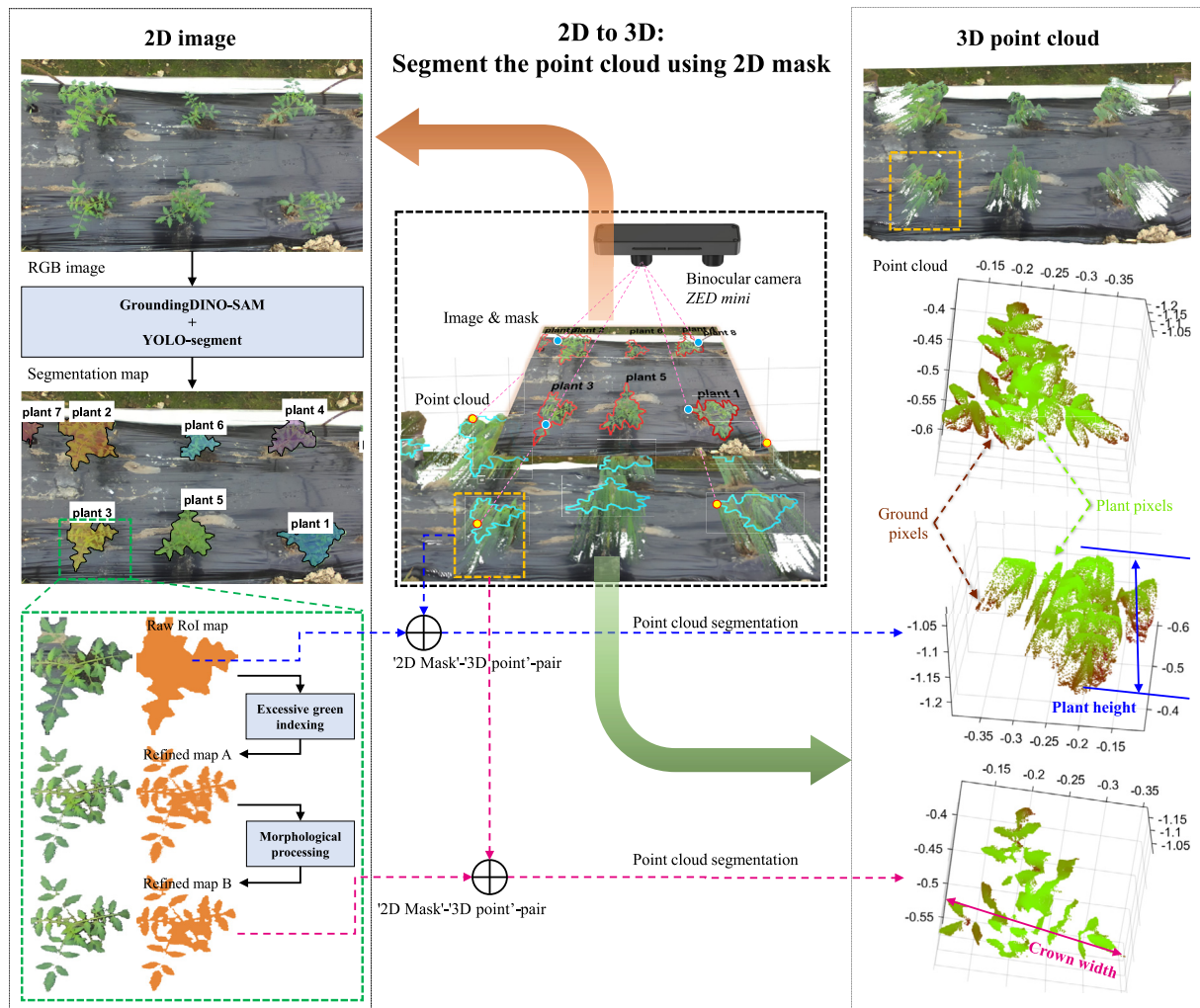


Fig. 1. Overview of the proposed tomato seedling 3D phenotyping method.

Table 1
Comparison of different 3D sensing devices.

Device type	Handheld 3D scanner	LiDAR	Binocular camera	RGB-D camera
Device model	Creality Otter Lite	FARO Focus Premium	ZED mini2	Intel D435i
Scanning mode	Inferred structured light and binocular imager sensors (360° scan)	Phase based laser ranging scanning	Two RGB cameras	Inferred projector and stereo imager; One RGB camera
Data type	3D point cloud	3D point cloud	3D point cloud, RGB image	RGB image, depth image, point cloud
Cost	\$607	\$59,250	\$399	\$334
Usage	Scan around the target	Fixed position	Instant imaging, can be mounted on automated robots	Instant imaging, can be mounted on automated robots
Time for a single capture	>30s(130s)	5–20 min	<1 s	<1 s
Time for post-processing	>60s(90s)	>1 h	<1 s	<1 s
Quality of output data	High-quality point cloud (complete)	High-quality point cloud (complete)	Relatively high-quality point cloud (single side), high image resolution (2 K)	Low-quality point cloud (single side), low image resolution (720p)
Data interface	Official software	Official software	Python SDK, official software	Python SDK, official software
SDK	N/A	N/A	python	python
Requirement on control unit	PC or mobile phone	PC	Edge computing device, PC	Edge computing device, PC

textual descriptions, while the image backbone used Swin-Transformer to extract multi-scale visual features from images. These text and image features were fused through a cross-modal feature enhancer. Then, a language-guided query selection module retrieved relevant queries from the enhanced image features, and these queries were processed and refined by a cross-modal decoder for detection and self-updating, ultimately generating predictions for object bounding boxes.

Segment Anything Model (SAM) (Kirillov et al., 2023) was an image segmentation model developed by Meta. It had robust zero-shot

generalization capabilities and incorporated a streaming memory module for real-time segmentation processing. The model dynamically accepted click or bounding box prompts at arbitrary image locations to define the target boundaries of the object, subsequently generating segmentation masks.

The combination of GD and SAM enabled zero-shot detection and segmentation of tomato plants (as shown in Fig. 2). The model versions used are Grounding DINO SwinB and ViT-L SAM. Notably, SAM had three different versions: ViT-B (358 MB), ViT-L (1.2GB), and ViT-H

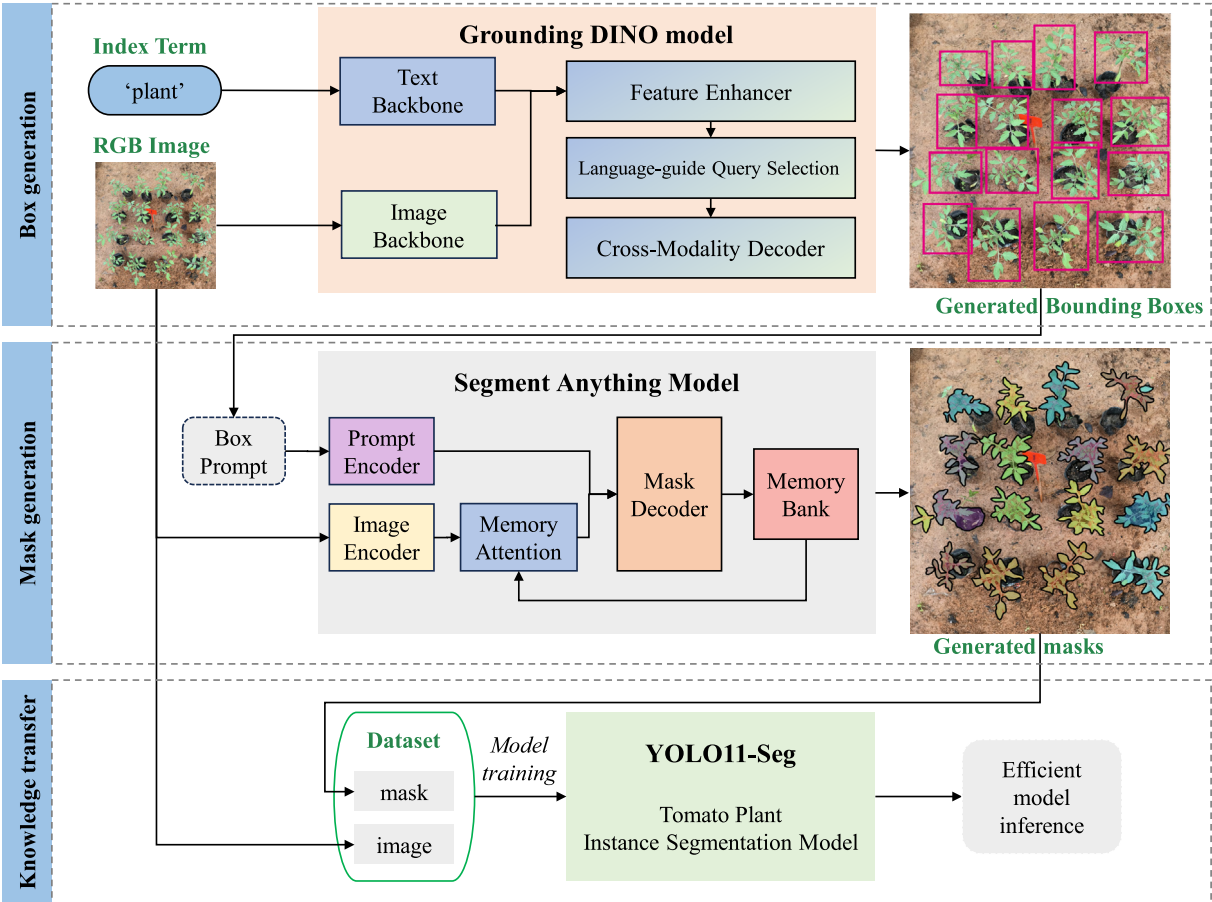


Fig. 2. Tomato plant instance segmentation method based on AI label generation and YOLO-segmentation.

(2.4GB). While the limited parameters of ViT-B hindered the accuracy of tomato plant segmentation, the excessive parameters of ViT-H imposed massive computational costs. Thus, ViT-L (308 million parameters) was selected paired with Grounding DINO SwinB. However, the massive parameter volume created substantial computational and memory demands, making it unsuitable for edge-device deployment. This necessitated developing a lightweight model to conduct the tomato plant segmentation task.

While GD + SAM could achieve tomato plant segmentation, the extensive parameterization of SAM made edge-device deployment impractical. To address this limitation, a transfer learning method was proposed to develop a lightweight tomato plant segmentation model. YOLO11-Segment (Khanam and Hussain, 2024), the latest instance segmentation model from Ultralytics, demonstrated significant improvements over previous YOLO iterations. It enhanced feature extraction capabilities that boost segmentation precision while maintaining real-time processing speeds. The model excelled in complex environments through optimized multi-scale feature fusion and achieved efficient instance segmentation with minimal computational overhead. With a compact 5.8 MB memory footprint, it is uniquely suited for resource-constrained embedded systems. Therefore, YOLO11-Segment was implemented as a lightweight solution for edge-deployable plant instance segmentation, as illustrated in Fig. 2.

2.3.2. Manual and AI generated labels

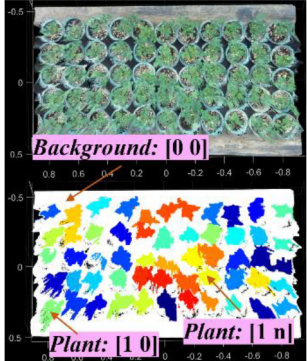
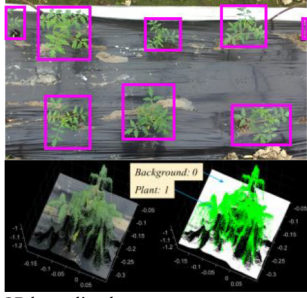
The final purpose of this study is 3D instance segmentation of tomato seedlings. The direct method is building an end-to-end 3D instance segmentation network. The required label is 3D point with class id and instance id, see Table 2, row 1, which is defined as strong label. Other labels which are simpler than this label '3D point with class id and instance id' are defined as weak labels. In Table 2, the smaller the complexity value, the weaker the label.

For obtaining the data at the row 5 in Table 2, as well as testing the time consumption of manual labeling, a thorough evaluation was conducted on 30 images, totally containing 1152 tomato plants. During label preparing, the manual revision involved correcting 146 plants, resulting in an averaged correction rate of 12.7%. This relatively low correction rate and minimal manual effort effectively underscore the practical value and affordability of the proposed method.

The format of different kind of labels are introduced below.

- (1) Labels for 3D instance segmentation: Assign each point in the 3D point cloud (including several plants) with a class id and an instance id, background = [00], plant = [1 N].
- (2) Labels for 3D semantic segmentation: Assign each point in the 3D point cloud (including only one plant) with a class id, background = 0, plant = 1.

Table 2
Label complexity and label quantity requirements of different AI models.

Method	Required labels	Required label type / quantity	Complexity of annotation ^a
3D instance segmentation	 3D point with class id and instance id	Strong label/Large quantity	1
2D object detection + 3D semantic segmentation	 2D bounding box; 3D point with class id	Strong label/Large quantity	2
2D instance segmentation + 2D-to-3D mapping	 2D mask	Weak label/Large quantity	3
2D object detection + SAM + 2D-to-3D mapping	2D bounding box	Weak label/Large quantity	4
GroundingDINO (with manual correction) + SAM + 2D-to-3D mapping	2D bounding box (15% correction)	Weak label/few-shot	5
GroundingDINO (no correction) + SAM + 2D-to-3D mapping	No label required	No label required/Zero-shot	6

^a Complexity of annotation are divided into six levels. '1' is the most complex. '6' is the simplest.

- (3) Labels for 2D object detection: 2D bounding box, YOLO-format txt files, where each line represents a tomato instance as: “class x y w h”, “x” and “y” were the normalized center coordinates of bounding box, “w” and “h” were the normalized width and height of the box.
- (4) Labels for instance segmentation: 2D mask, YOLO-format txt files, where each line represents a tomato instance as: “class x1 y1 x2 y2 ... xn yn”, where “class” denoted the category, and “x1” and “y1” were the normalized horizontal and vertical coordinates of the first contour point, followed sequentially by “x2, y2”, “x3, y3” etc.

For 3D processing, the instance segmentation dataset includes 209 raw point-cloud files. Then, the semantic segmentation datasets are extracted from the raw point clouds, obtaining 4089 point-cloud files (each of them contains 1 plant instance). Both the mentioned datasets were divided into training (60%), validation (20%) and testing (20%).

For 2D processing, 500 tomato images raw images are used in both the object detection and instance segmentation datasets. Both the mentioned datasets were divided into training (60%), validation (20%) and testing (20%). It must be highlighted that, here the labels in training and validation datasets are obtained by Grounding DINO, manual correction on bounding box and SAM for reducing the manual cost, while the labels in testing sub-datasets are all pure manual annotations to better evaluating the performance of the trained models.

2.4. Coordinate mapping between 2D images and 3D point clouds

The fundamental mechanism of coordinate mapping establishes a correspondence between 2D pixel coordinates (u, v) in images and 3D point cloud coordinates (X, Y, Z) by leveraging the intrinsic and extrinsic parameters of binocular cameras. The pinhole imaging model of a monocular camera establishes a unique mapping between points in physical space and their corresponding points in the image. The specific correspondences are as follows:

$$Z \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \quad (1)$$

where u, v represents the pixel coordinates of the image, f_x, f_y, c_x, c_y represents the intrinsic parameters of the camera, and X, Y, Z represents the coordinates in physical space. For stereo vision systems using binocular cameras, two images are captured from different viewpoints. To accurately compute depth information for each pixel, these images must first undergo distortion correction and stereo rectification:

$$u_{corrected} = u(1 + k_1 r^2 + k_2 r^4) + 2p_1 uv + p_2(r^2 + 2u^2) \quad (2)$$

$$v_{corrected} = y(1 + k_1 r^2 + k_2 r^4) + 2p_2 uv + p_1(r^2 + 2v^2) \quad (3)$$

where u and v refer to the pixel coordinates in the original imaging of the binocular camera, k_1, k_2, p_1, p_2 refer to the distortion coefficients of the camera and $r^2 = u^2 + v^2$. Subsequently, the correct depth data, combined with the intrinsic parameters of the camera, enables the calculation of the spatial coordinates (X, Y, Z) corresponding to any given pixel (u, v).

$$X_L = \frac{b \cdot (u_L - u_0)}{d} \quad (4)$$

$$Y_L = \frac{b \cdot (v_L - v_0)}{d} \quad (5)$$

$$Z_L = \frac{f \cdot b}{d} \quad (6)$$

where b represents the distance between the two lenses of the binocular camera, u_L and v_L denote the coordinates of the optical center of the left lens, u_0 and v_0 denote the coordinates of the x-axis and y-axis on the image, d represents the disparity value corresponding to the pixel, f represents the focal length of the camera.

Once tomato plant masks were generated by the YOLO11-Segment segmentation model, the masked image regions were spatially mapped to the 3D point cloud, enabling the extraction of corresponding plant-specific point subsets. The workflow is shown in Fig. 3. The codes to help reproduce the 2D-to-3D segmentation method in this study is published on GitHub (<https://github.com/L-Zhou17/2D-to-3D-segmentation>).

2.5. Extraction of phenotypic parameters

2.5.1. Calculation of crown width

Crown width, defined as the maximum diameter between the widest points of a plant canopy during growth, served as a critical phenotypic parameter for assessing plant development. It reflected sunlight and water absorption capacity, influenced resistance to pests and environmental stress, and enabled crop yield estimation (Poudyal et al., 2019). Traditional manual measurement suffered from inefficiency, subjective errors and low precision. Spatial data from 3D point clouds was employed for batch-processing of crown width calculations. To address the computational complexity of pairwise point distance evaluation and avoid directional bias (e.g., neglecting diagonal maxima when measuring y-axis distances alone), the rotating calipers algorithm was implemented as follows: After isolating individual plant point clouds, preprocessing and coordinate normalization, all points were vertically projected onto the x-y plane. Canopy vertices were connected to form a convex hull, and the rotating calipers method determined the maximum crown width (Fig. 4).

The process initialized antipodal point pairs (the farthest points along a convex hull axis), clamped the hull between parallel caliper lines, then rotated these lines clockwise to sequentially measure distances between updated antipodal pairs. The maximum recorded distance after a full 360° rotation defined the convex hull's maximum diameter, corresponding to the plant's crown width. The computational formula for crown width was expressed as:

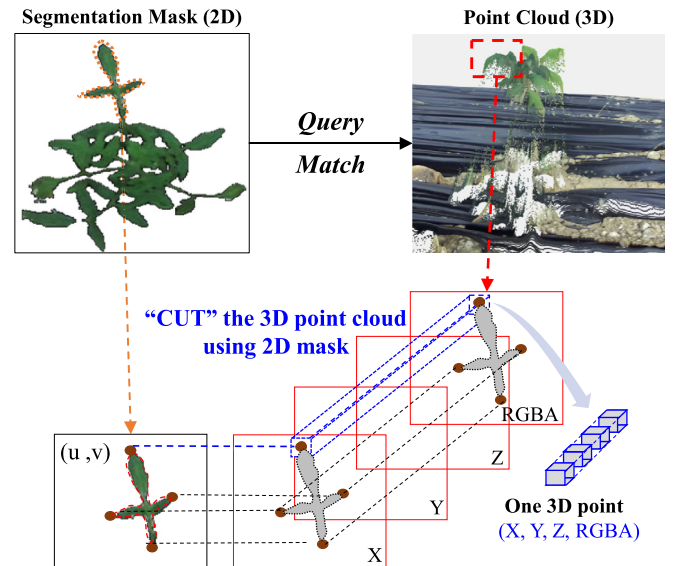


Fig. 3. Tomato plant seedlings 3D point cloud segmentation using a 2D mask produced by the YOLO model.

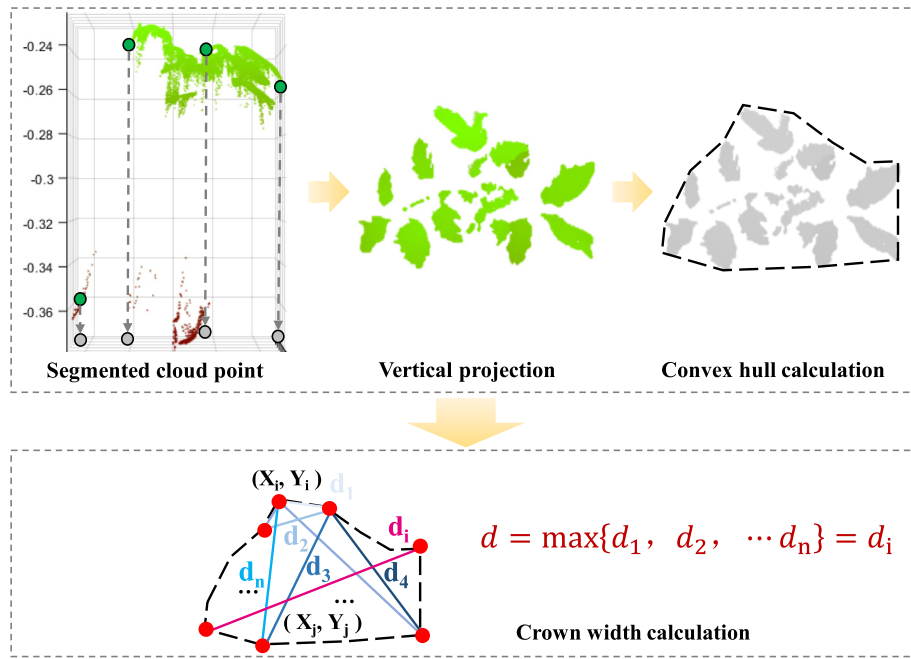


Fig. 4. Process of tomato plant crown width calculation method.

$$d = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (7)$$

where d is the calculated crown width. (x_i, y_i) , (x_j, y_j) are two selected points.

2.5.2. Calculation of height

Plant height was a pivotal phenotypic parameter in botanical research, integrally capturing the influence of genetic predispositions, environmental conditions, and agricultural practices. Continuous monitoring of greenhouse tomato plant height enabled quantitative assessment of seedling vigor, photosynthetic efficiency, and stress resilience while informing precision irrigation and nutrient management protocols.

- (1) Height calculation for tomato seedlings with a sparse canopy and relatively wide spacing between rows and columns, which are directly planted in the ground without pots. The experimental height quantification workflow proceeded as follows: For an isolated tomato plant point cloud model, following pre-processing and coordinate normalization, the top 30 points with ascending Z-axis coordinates (representing the highest spatial positions) and the bottom 30 points with descending Z-axis coordinates (representing the lowest positions) were selected. Their respective Z-coordinate means were computed as Z_{max} (peak elevation) and Z_{min} (base elevation) to enhance statistical robustness against outliers. The absolute difference between these Z-values defined the plant height. The above-mentioned computational process was visualized in Fig. 5(d).
- (2) Height calculation for tomato seedlings or other plant types possess a dense canopy, which are planted in pot with relatively narrow spacing between rows and columns. It should be highlighted that, for some plant objects, top-view sensing and the corresponding segmentation results may not cover the points of bottom part of the plant or the ground. The calculation of the lowest value for plants will result in significant deviation. Hence, a new method and some required hardware are presented in Fig. 5(a), (b) and (c). A chessboard pattern was used

as a reference plane, for calculating the z-value of the ground in a more accurate way. The calculated plant height value H_l involves the height of the pot (or cultivation substrate). Commonly, modern facility greenhouse adopts standardized planting mode. The height of the pot is commonly a fixed value in one greenhouse. Therefore, the actual plant height could be calculated by subtracting the fixed pot height from the height value of H_l .

3. Results

3.1. Tomato seedling detection and segmentation

3.1.1. Bounding boxes calculation

This study tried to utilize the popular large vision model GD to generate bounding boxes of tomato plants, which were further used as the prompt information of the SAM for segmentation. The raw output of GD included some issues, such as missed inspections and duplicate inspections. Some manual correction is required, see Section 2.3.2. The corrected box information was further used for YOLO detection model training, obtaining an accurate box prompt for the SAM. Typical deep learning object detection methods were used and compared for tomato seedling detection, see Table 3.

For training the YOLO models, the epoch was set as 100, the resolution was set as 640, the learning rate and the early-stop procedure used the default configuration of the Ultralytics package. The model with the highest mAP_{50} value on the validation dataset was saved as 'best.pt' for further analysis. As shown in the table above, high detection performance was achieved across all evaluated object detection models, with mAP_{50} exceeding 96%. Notably, GD + YOLOv11 achieved the highest mAP_{50} of 97.2%. Furthermore, GD + YOLOv11 demonstrated significant precision improvements over GD + YOLOv5, v8, v9 and v10 with gains of 2.2%, 1.7%, 2.3% and 0.8% respectively. The statistical significance testing was performed between YOLO11 and other different models. The p -values concerning YOLOv5, YOLOv9, and YOLOv10 were all less than 0.05, indicating the significance of the YOLOv11. That value for YOLOv8 is 0.14, indicating that the

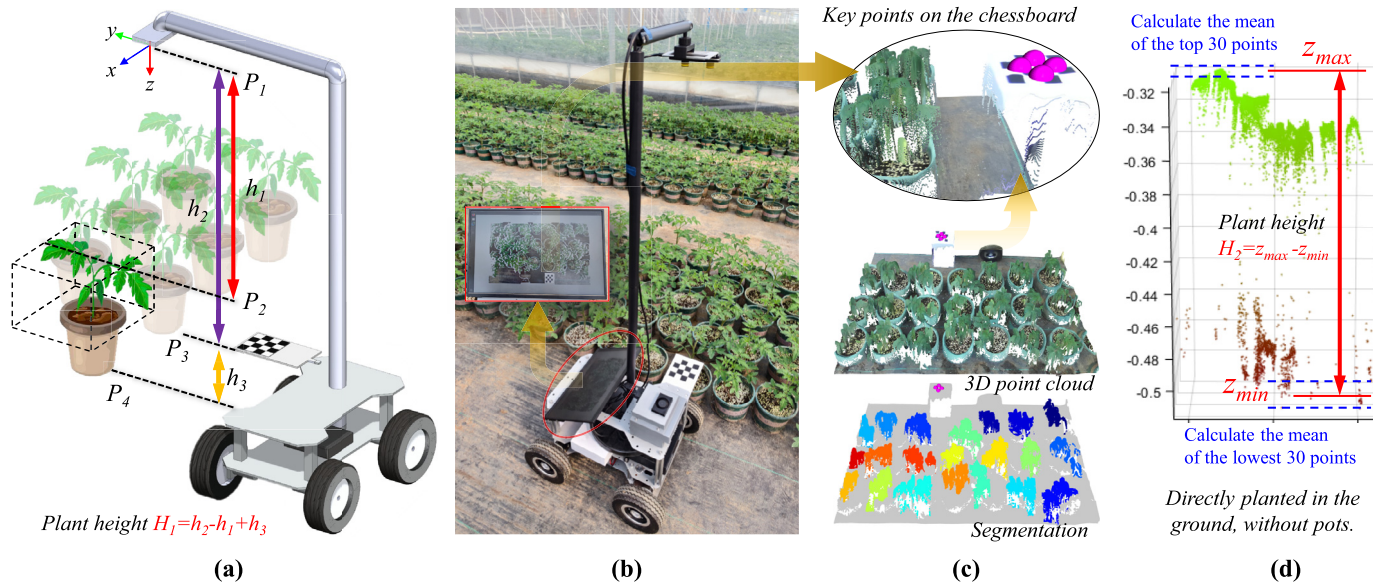


Fig. 5. Tomato plant height calculation method. (a) Diagrammatic sketch of the designed device and method for height calculation. H_1 denotes the height of plant grown in pot calculated from the ground. (b) Real picture of the designed device, (c) result visualization of plant 3D segmentation and reference board detection, (d) single plant height calculation. H_2 denotes the height of plant direct grown in ground.

Table 3
Performance of different tomato plant detection models.

Model	Performance				
	P	R	mAP ₅₀	mAP _{50–95}	Size (Mb)
GD + YOLOv5	0.934	0.936	0.968	0.792	5.3
GD + YOLOv8	0.939	0.934	0.969	0.797	6.2
GD + YOLOv9	0.933	0.921	0.964	0.792	4.7
GD + YOLOv10	0.948	0.918	0.965	0.790	5.6
GD + YOLOv11	0.956	0.935	0.972	0.798	5.4

performance of these two segmentation models is similar. Based on the above-mentioned comparative analysis, GD + YOLOv11 was selected as the best tomato seeding object detection model for this study.

3.1.2. Segmentation masks generation

Based on the above-mentioned plant object detection results (Precision = 0.956 and mAP₅₀ = 0.972), this high-quality box information was further adopted to assist the SAM in getting segmentation masks of tomato plants. During this procedure, no manual annotation work or correction operation was involved. Plant instance segmentation results are listed in Table 4.

3.1.3. Mask processing and optimization

The reduced feature map resolution caused by the lightweight detection head of the YOLO model (which prioritizes speed over accuracy), repeated use of down-sampling methods in the backbone network, and mosaic data augmentation techniques lead to blurred mask edges that fail to precisely align with target contours. As shown in Fig. 6(a), the initial segmentation results of the YOLO model contain

Table 4
Performance of different tomato plant segmentation models.

Model	Performance				
	P	R	mAP ₅₀	mAP _{50–95}	Size (Mb)
SAM + YOLOv8-segment	0.943	0.926	0.955	0.689	6.7
SAM + YOLOv11-segment	0.945	0.934	0.960	0.692	5.8

substantial non-target debris such as plastic mulch and gravel. A post-processing method based on excess green index (ExG) feature segmentation was implemented to address these limitations (Han et al., 2023). The ExG value for each pixel in RGB images was calculated as:

$$V_{\text{exg}} = 2G - p(R + B) \quad (8)$$

where R , G , and B represent the red, green, and blue channel values, respectively. Furthermore, the formula operated under the precondition that $G > R$ and $G > B$ (where $p = 1.2$). Pixels yielding a computed value greater than zero were classified as foreground; all others were considered background. Precise plant extraction was achieved, as demonstrated in Fig. 6(b). Finally, morphological erosion was employed to eliminate residual isolated noise points after ExG feature segmentation. As illustrated in Fig. 6(c), this operation successfully removes pseudo-edge pixel noise points while preserving critical edge details like serrated leaf margins and stem structures. The refined masks now accurately align with the authentic morphological characteristics of the plants. By the fusion of YOLO segmentation and ExG feature segmentation, a more precise mask could be obtained for conducting subsequent plant 3D parameters calculation steps.

3.2. Tomato seeding point cloud segmentation

Processed 2D masks of tomato plants and spatial coordinate mapping were used to segment 3D point clouds. Some typical examples of point cloud segmentation results of a single tomato plant are shown in Fig. 7. Objects such as flowerpots and sandy soil are visible in the pre-segmentation 2D images. After applying the proposed point cloud segmentation method, only the expected 3D spatial information of plant remains in the point cloud. This spatial data could then be used to extract specific phenotypic parameters of the tomato plant.

3.3. Results of phenotypic parameter extraction

Conventional 2D image processing methods commonly calculate plant size by analyzing pixel quantities of reference objects in binary mask images, which is contrasted with the phenotyping parameter extraction method proposed by this experiment, and the result is shown in Table 5.

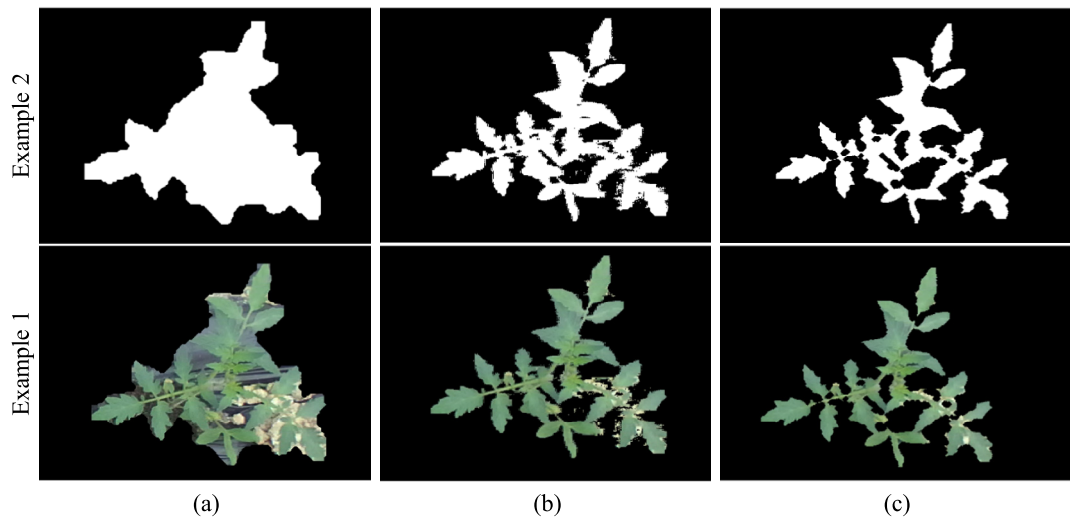


Fig. 6. Results of tomato plant segmentation using YOLO coupled with mask optimization methods. (a) YOLO segmentation output ($Mask_{raw}$), (b) $Mask_{raw}$ processed by ExG feature segmentation ($Mask_{ExG}$), (c) $Mask_{ExG}$ processed by morphological operation.

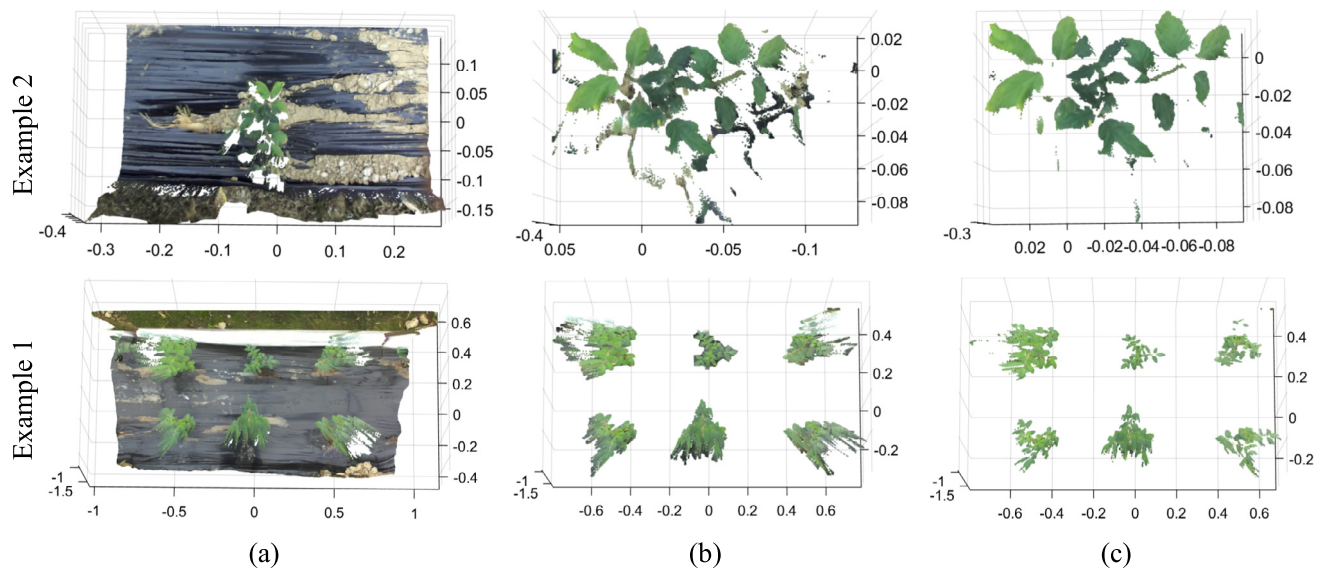


Fig. 7. Examples of tomato seedling 3D point cloud segmentation results based on AI segmentation and 2D-3D-mapping. (a) Raw point cloud data, (b) segmented points based on raw YOLO masks, (c) segmented points based on refined masks.

Table 5
Performance of different methods for tomato crown width and plant height calculation.

Method	Plant height ^a		Crown width	
	R ²	RMSE	R ²	RMSE
2D image	N/A	N/A	0.79	2.9
3D point cloud	0.93	1.7	0.95	1.0

^a It is hard to measure plant height using only 2D images. Therefore, the corresponding units are filled with N/A.

While extracting phenotypic parameters using 2D images, plant height cannot be directly determined due to the lack of depth information. In contrast, 3D point cloud data effectively addresses this limitation (Fig. 8). Parameters extracted from point cloud data exhibit higher accuracy compared to those obtained from traditional 2D methods.

The comparative analysis between the parameters calculated by the proposed method and manual measurements of crown width and plant height is presented in Fig. 9. Manual measurements were conducted by using rulers with a resolution of 0.1 cm.

For evaluation of plant with a sparse canopy, 100 tomato samples were selected (Fig. 9 (a) and (b)), reference chessboard in not necessary here.

Statistical analysis revealed significant linear correlations between the extracted and measured values for both parameters. The AI vision-based crown width measurements demonstrated strong agreement with manually measured data, showing a Coefficient of Determination (R^2) of 0.95 and Root Mean Square Error (RMSE) of 1.0 cm. Plant height extraction achieved an R^2 value of 0.93 with an RMSE of 1.7 cm. The superior accuracy in crown width estimation compared to plant height measurement may be attributed to the inherent challenges in plant vertical dimension, which is more susceptible to leaf occlusion and imaging angle. Moreover, the error inherent in selecting ground control points as elevation references substantially compromised the precision of plant height estimation. Moreover, the study included tests on very young tomato seedlings (e.g., at the two-true-leaf stage). Image segmentation and point cloud reconstruction for these seedlings presented significant challenges, resulting in substantial computational errors for phenotypic parameters.

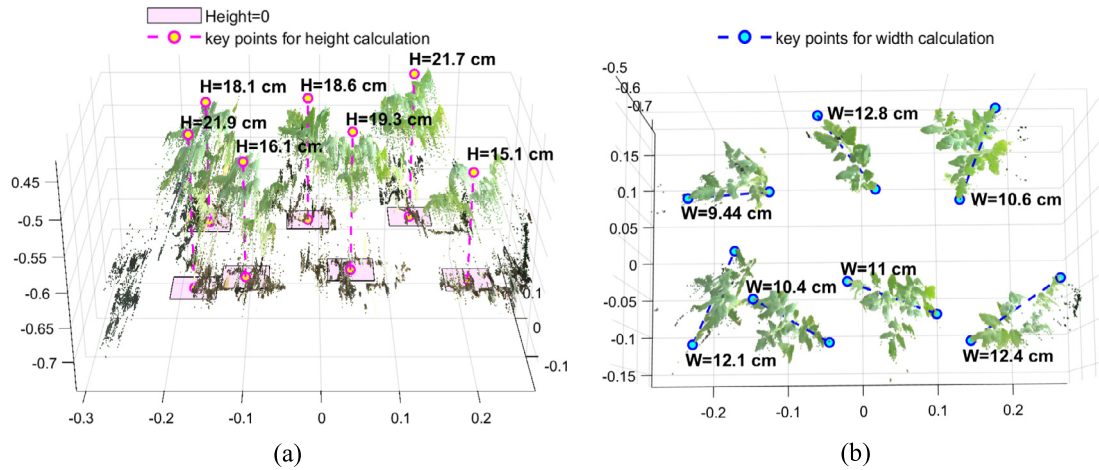


Fig. 8. Tomato seedling phenotypic parameters determination using a segmented point cloud. (a) plant height calculation, (b) crown width calculation.

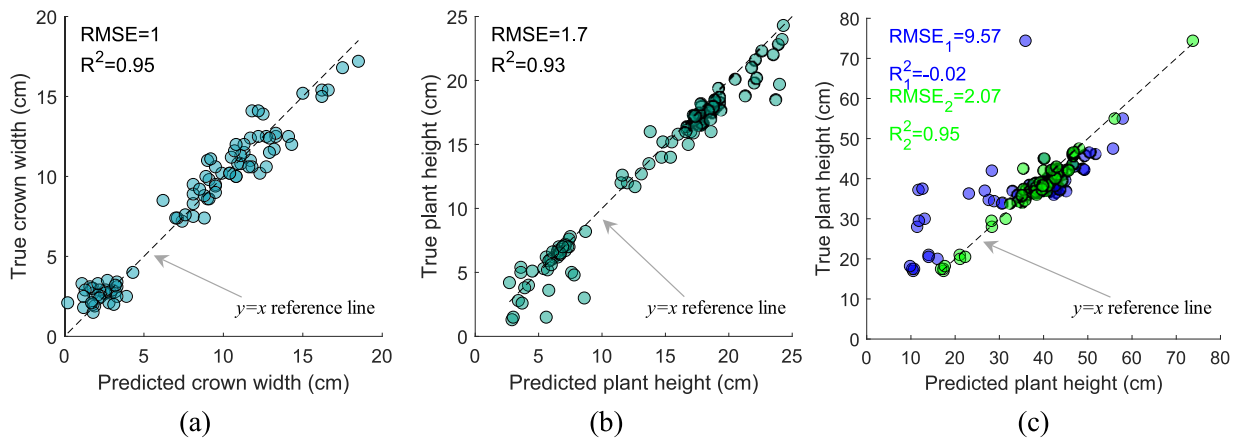


Fig. 9. Scatter plots of tomato plant phenotypic parameter extraction results. (a) crown width of the tomato plant, (b) height of the tomato plant with a sparse canopy, (c) height calculation of an extra dataset (involving densely arranged tomato plant during different growth stage, a lemon tree, eggplant seedlings and passion fruit seedlings). Note: green scatters are the results calculated with reference board, blue scatters are calculated without reference board.

Notably, both parameters exhibited satisfactory prediction accuracy with R^2 values exceeding 0.9 and RMSEs below 2 cm. These results validated the effectiveness of the proposed 2D mask-based 3D point cloud segmentation method, demonstrating its capability to meet the precision requirements for plant phenotyping analysis.

For evaluation of tomato at different stage and other plant type with a dense canopy, another 60 plants were analyzed (Fig. 9(c)). Green scatters are the results calculated with reference board. Blue scatters are calculated without reference board. From the top-view, the camera cannot obtain the bottom part of these plants and cannot get the ground pixel under the canopy. Therefore, without the reference board, it is hard for the AI framework to find the ground plane. The RMSE of height calculation is very high (9.57 cm). Conversely, for analyzing such plant samples, height calculation using the reference board improved the accuracy, with only an RMSE of 2 cm.

3.4. Segmentation accuracy and computational efficiency on edge computing device

The accuracy and computational efficiencies of the proposed 3D segmentation method, as well as the compared existing methods, were summarized in Table 6. To conduct a fair comparison, accuracy indicators that can be evaluated for multiple segmentation methods are listed and discussed. Direct 3D segmentation method (Boogaard et al., 2023;

Xie et al., 2025), PointNet++ with 3D instance embedding head, only achieved a very low Average Precision value. The reason can be trace back to the raw 3D point cloud data, see Fig. 1. In some image-point cloud-pairs, plants only cover a small region, while most of the points belong to the background. The point cloud downsampling operation in the model will lose many plant points. YOLO + PointNet++ 3D semantic segmentation means: use the YOLO box to 'cut' the raw point cloud, obtaining sub-point cloud including only one plant and some background points; then use PointNet++ semantic segmentation to discriminate the plant and the background. Relatively high performance was observed. However, the problem of losing some points of the plant still exists. As for the proposed 2D-to-3D Segmentation method, it segmented the plant and provided the edge contour of the plant fewer inner points were lost, therefore the average precision value was high. But its average recall is lower than YOLO + PointNet++ 3D semantic segmentation.

The time (on edge device) for segment the 3D point cloud of one plant was discussed at first. The commonly used PointNet 3D semantic segmentation spent 1.8 s–15.7 s per instance. The more 3D points a single plant has, the longer the segmentation time. While the proposed 2D-to-3D Segmentation method achieved about 0.14 s per instance for segmentation in average. It increases the segmentation speed by more than ten times. And its time consumption is almost unrelated to the number of 3D points per plant. Then, the total time (T_{total}) are analyzed for

Table 6

Performance comparison of tomato 3D segmentation methods on edge computing device (NVIDIA Jetson Orin NX).

Method	Time cost				Accuracy indicators	
	Plant number in one image ^a	YOLO (s/f) ^b	ExG (s/f) ^c	3D segment (s/it) ^d	Average Precision	Average Recall
2D-to-3D Segmentation	3	0.157	0.988	0.142	0.949	0.928
	15	0.184	1.025	0.145		
	20	0.123	1.020	0.145		
	50	0.177	1.026	0.144		
YOLO + PointNet++ 3D semantic segmentation	50 (200 kb/it)	0.186	1.052	1.760	0.896	0.958
	20 (500 kb/it)	0.172	1.042	6.200		
	3 (3000 kb/it)	0.151	0.997	15.710		
PointNet++ with 3D instance embedding head	N/A				0.132	0.6352

^a kb/it means kilobits per instance. 50 (200 kb/it) means that in one ‘image-3D point cloud pair’ covering about 50 instances, the size the point cloud file of one single plant is about 200 kb.

^b s/f denotes second per frame. One ‘image-3D point cloud pair’ requires YOLO to inference once.

^c ExG need to be performed once on one image.

processing one image covering N plants. It could be calculated by $T_{\text{total}} = 1 \times (T_{\text{YOLO}} + T_{\text{ExG}}) + N \times T_{\text{3D Segment}} + T_{\text{Width and height calculation}}$. Hence, T_{total} for processing an ‘image-3D point cloud pair’ covering 20 plants was about 6.7 s based on the proposed 2D-to-3D Segmentation method. While T_{total} for processing similar data using YOLO + PointNet++ 3D semantic segmentation was higher than 120 s. The proposed new method showed obvious superiority in data processing efficiency.

4. Discussion

The seedling-stage cultivation of tomatoes significantly influences subsequent plant growth and fruit yield, underscoring the critical importance of seedling-stage tomato selection.

This study utilized the ZED mini2 stereoscopic camera to simultaneously capture 2D images and 3D point cloud data of tomato seedlings. Then, GD + SAM was employed to achieve plant instance segmentation under few-shot conditions, with the segmentation model subsequently transferred to the more computationally efficient YOLOv11-seg model. The ExG algorithm was further applied to optimize the segmentation masks through post-processing, while a 2D-3D coordinate mapping approach enabled lightweight point cloud segmentation. Finally, plant height and crown width of tomato seedlings were calculated from the processed plant point cloud data. The innovative works, contributions, as well as limitations are summarized and discussed as follows.

4.1. Advantages of this study

(1) Advantages in 3D point cloud collection and segmentation.

Current studies predominantly focused on high-precision reconstruction of plant point clouds (usually collected by a Lidar or 3D scanner) followed by direct fine segmentation of 3D cloud points to extract phenotypic parameters (Dong et al., 2025; Li et al., 2025). This kind of approach delivered high accuracy, however, such scanning devices usually could not provide a matched RGB image. Therefore, a complex 3D deep learning method should be used to process point clouds directly, which entails complex processing pipelines and labor-intensive 3D data annotation (Jin et al., 2020). In the presented study, real-time 3D point cloud data collection could be achieved by the binocular camera. This requires fewer data collection operations and time. Then, a 2D mask-driven 3D point cloud segmentation methodology was proposed, circumventing direct manipulation of complex 3D data to achieve low-cost, high-efficiency, and lightweight point cloud segmentation. The proposed method realized similar segmentation accuracy than PointNet++ based method, but the inference speed is more than ten times faster than the latter.

(2) Innovation in 2D instance segmentation.

Conventional deep learning image segmentation relied heavily on manual annotation. It is a time-consuming and laborious process with low efficiency (Yue et al., 2023). Recent studies have employed models like GD and SAM to achieve instance segmentation of agricultural objects, but their direct deployment demanded substantial computational resources and high-performance hardware. This study exported annotation results from GD and SAM to compile training datasets, then migrated knowledge via transfer learning to a lighter, faster YOLO11-segment model. This methodology not only significantly reduced annotation time but also delivered efficient and lightweight plant instance segmentation to avoid the usage of high-cost and large-volume computing devices.

(3) Rapid transferability.

The proposed 3D plant segmentation and phenotyping method is easy to be expanded its usage. Based on the program published by the authors, the potential users just need to change the YOLO weight file (.pt) to improve the scope of this method, enabling the AI framework to handle new samples in the extra dataset. For tomato samples planted in-pot with different planting density, satisfactory results could be observed by fine-tuning the YOLO-segment model, see Fig. 10(a). It is also easy to transfer this method for other plant type like lemon tree, eggplant seedlings and passion fruit seedlings, see Fig. 10(b) and (c). VLMs could be used to generate labels for more crop types by the text prompt ‘plant’ or ‘crop’. The generated labels could be further used to train a general light-weight plant segmentation model covering multiple plant varieties and species, making full use of strong generalization capability with minimal task-specific training. In future study, the proposed method could also be used for a broader type of agricultural objects. According to the existing publications, VLMs could realize a better segmentation performance on fruit targets using their zero-shot generalization capabilities. By modifying the proposed AI framework, it is possible to simultaneously perform fruit size estimation, 3D posture determination and positioning tasks.

(4) Improvements on phenotypic parameter calculation.

When measuring plant height and crown width parameters, conventional 2D imaging methods can only obtain relative dimensions in pixels, requiring reference objects for physical scale conversion. Moreover, for height measurement, lateral-view photography is essential, which not only increases time costs but also introduces inherent difficulties in reference marker positioning caused by stem occlusion effects. While alternative approaches based on top-view 3D point clouds

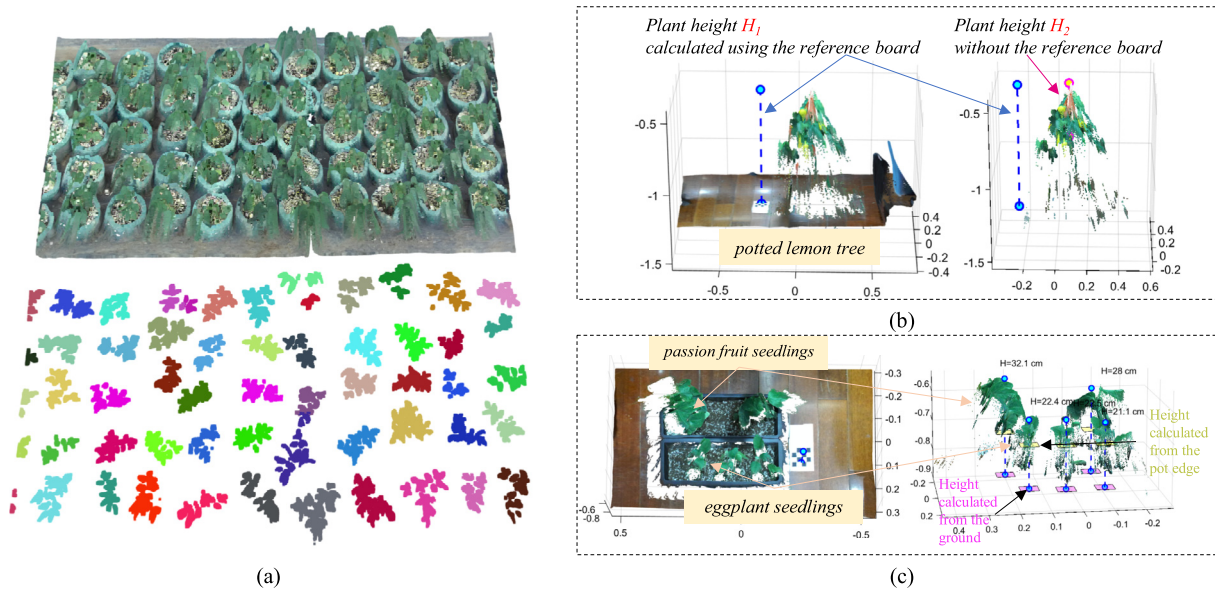


Fig. 10. Results of 3D segmentation on an extra tomato dataset and the usage of proposed method on new plant type. (a) Segmentation of ‘image-point cloud-pair’ covering 50 tomato plants. (b) Proposed method for lemon tree height calculation. (c) Proposed method for eggplant seedlings and passion fruit seedlings height calculation. Note: in (b) and (c), height calculation from the ground, edge of the pot and from the lowest point in the segmented point cloud are compared.

eliminate the need for multi-perspective imaging, direct point cloud segmentation captures only the canopy structure of plants, necessitating additional computational methods to determine the ground reference point for height measurement (Du et al., 2023), and AI models processing raw 3D point clouds directly involve significantly more complex and time-consuming data annotation during training (Matsuura et al., 2023).

The proposed method utilizes single-perspective imaging, achieving accurate spatial localization of target plants. The original YOLO segmentation combined with 2D–3D mapping retained the growth substrate background. For processing plant with a sparse canopy, this brought about advantages instead of problems for plant height calculation, as the substrate served as a reference plane of height = 0. By utilizing ground point clouds from localized areas near detected plants instead of large-scale surface fitting, such method (in Fig. 5(c)) does not require the reference board in Fig. 5(a). However, for those plants with a dense canopy (in Fig. 10(b)), the part below the canopy could not be observed from the top-view. Without the reference board in Fig. 5(a), only the canopy thickness could be measured. With the help of the reference board, accurate depth value of the ground could be determined. It should be mentioned that, in modern greenhouse, usually standardized planting is adopted using a uniform pot and soil thickness, allowing users to choose the zero position for calculating plant height based on their specific needs. In Fig. 10(c), height measurement from the ground and that from the pot edge was compared.

(5) Affordable costs concerning time, usage, equipment, professional knowledge learning.

Table 1 summarized typical existing 3D techniques. ZED mini2 stereoscopic camera shows superiority about the device cost, data collection speed, feasibility of deployment on edge computing devices, control interfaces and other key factors. Table 2 compares the complexity of preparing different types of annotations. On the premise of ensuring modeling accuracy, a method with less annotation workload and simpler process was chosen. Table 6 shows that the proposed 2D-to-3D mapping method increases the segmentation speed by more than ten times when compared to those methods directly using AI model for 3D point processing. Moreover, this test is conducted on an edge computing device, which prove the practical application value of this method. Overall, the proposed method is easy to learn, use, and

understand. It is relatively friendly to agricultural researchers, managers, breeding personnel and so on, without professional computer science-related knowledge. Besides, our published codes could help the readers reproduce the presented method faster and simpler.

4.2. Practical values

The proposed methodology demonstrates tripartite application value in agricultural practice. First, featuring a memory-efficient and computationally lightweight design, the proposed tomato seedling 3D phenotyping method could be directly deployed on edge devices in field environments or greenhouses, thereby achieving dynamic plant monitoring and real-time data processing. Second, it could help precisely analyze seedling-stage tomato growth status, providing data foundations for intelligent irrigation and fertilization decision systems. Third, smart farming systems are widely adopted for automated management in controlled-environment tomato cultivation. Failure to promptly remove inferior seedlings allows these plants to persistently occupy valuable cultivation space, reduce water and fertilizer use efficiency, and negatively impact final fruit yield. This study could enable the selection of superior tomato seedlings in controlled-environment cultivation by extracting plant height and crown width data during the seedling stage to establish a growth potential evaluation model. Besides, based on the designed method showed in Fig. 5(a) and the results in Fig. 10(b), the center position (X_c , Y_c , Z_c) of the segmented plant is determined. By modifying the Z_c according to the height of the pot, the center coordinates of the pot could be calculated. It provides target operation point for smart agricultural machinery, for further conducting automated capture, movement, and handling of potted seedlings.

4.3. Limitations, technical constraints, and future perspectives

Current popular AI models are commonly trained based on the dataset of objects in daily life. Therefore, they were good at identifying those objects. However, these models exhibited critical gaps in agricultural-specific training datasets, requiring a three-stage compensatory workflow for object extraction in the field of agriculture. i) initial object detection of plant targets, ii) subsequent manual

annotation refinement, and iii) ultimately the conversion to standardized box prompts. The above-mentioned factors make it difficult to achieve complete zero-shot learning in this study. Encouragingly, some cases show that VLM (e.g., the combination of GD and SAM) could realize complete zero-shot learning for fruit object detection (Giménez-Gallego et al., 2024). Hence, in the future, research could be conducted on how to expand the recognition scope of VLM or be dedicated to the development of large agricultural models. In addition, other creative ideas could be explored. For example, providing effective prompt information for the SAM. In Bhattarai's study (Bhattarai and Karkee, 2022), a weakly-supervised deep learning model was used for apple flower and fruit segmentation. The local maximum values in the attention map indicate the locations of the fruits and flowers. This information is also promising to be the prompt of the SAM to achieve fruit or flower instance segmentation. Such method could also be transferred to tomato plant segmentation task in the future.

As a fact, there are also some limitations of the designed 3D phenotyping methods. Errors in plant height and crown width could be caused by different factors. Due to the performance of the camera itself, seedlings with too small in size are not easy to be detected by the segmentation model, and also hard to reconstruct to get a 3D point cloud, see Fig. 11(a). Another reason is that, the erroneous samples during 3D plant phenotyping are mainly caused by some inaccurate YOLO-generated masks, see Fig. 11(b). Therefore, future study should focus on how to get more accurate masks for tomato plants. Another problem is that, the binocular camera appears to produce noticeable point cloud drift, see Fig. 8 and Fig. 11. This may influence the accuracy of phenotypic traits measurement.

Based on the analysis results of this study, the problem of point cloud drift mainly occurs at the location of sudden changes in depth values (target object edge contour). A possible solution is that, a confidence value of each point could be assigned. The closer the point is to the

edge contour, the lower the confidence value. Subsequent analysis could be conducted based on confidence weighted point cloud data, for improving the measurement accuracy of other relevant traits (e.g., leaf inclination angle, internode distance, leaf area, and stem diameter).

The issue of plant occlusion in tomato phenotype analysis is discussed from two aspects.

- (1) In modern tomato cultivation, standardized planting and standardized data collection are the current development trends.

For scientific research purposes (e.g., stress resistance test (Ma et al., 2025)), as well as for selling cultivated tomato seedlings, the seedlings in these stages are usually in the 5–7 true leaf stage, or earlier (Sangeetha et al., 2020; Yadav et al., 2025). The crown width is commonly lower than 25 cm. For some mini tomato varieties, the size at mature stage is also small (width < 25 cm). Traditional planting bases or modern facility greenhouses usually control row spacing (> 50 cm) and plant spacing (> 35 cm). Such planting mode not only avoids nutrient competition between plants, but also avoids the problem of plant overlap and obstruction. Moreover, for tomato breeding or stress resistance test, the planting spacing will be larger, providing convenience for single plant monitoring and observation.

- (2) The high planting density makes it difficult to conduct individual plant analysis.

For targets with relatively regular shapes, such as apples, it is possible to recognize them and use algorithms to restore the occluded parts when individuals are mutually occluded. However, the irregular shape of tomato canopy makes it hard to distinguish individual plants when they are occluded. Even human experts cannot provide accurate and

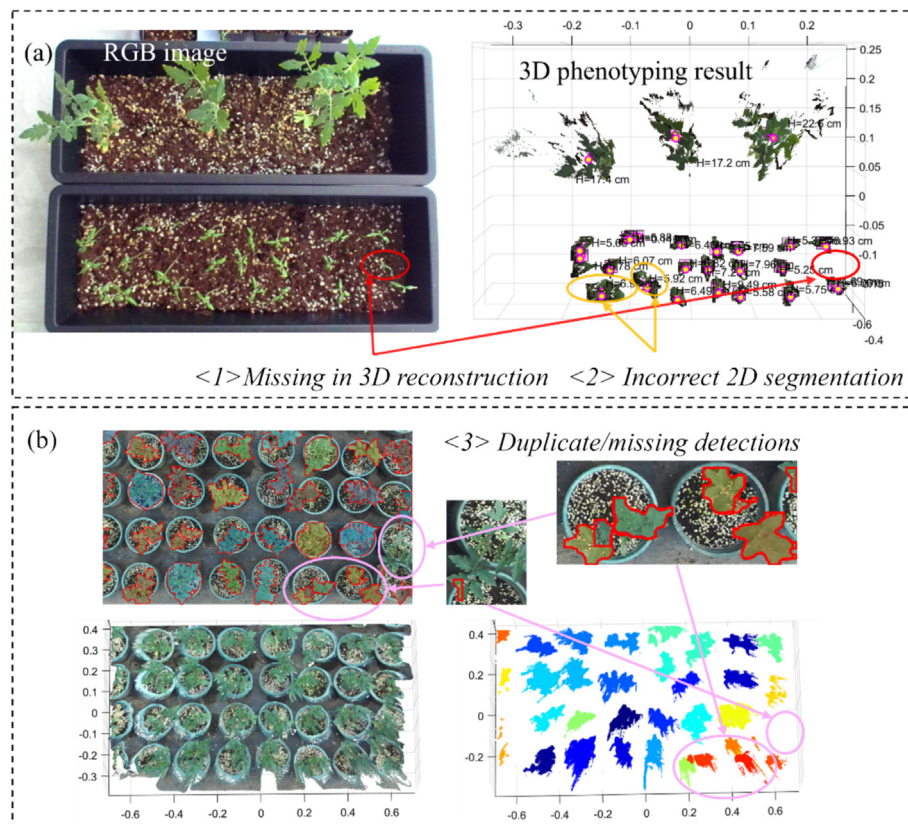


Fig. 11. Examples of some failure cases for tomato seedling segmentation. (a) Example 1, (b) Example 2.

reliable labels for AI modeling. Therefore, in this study, plant spacing and row spacing are controlled. For plant sample in traditional greenhouse (see Fig. 7), the spacing is about 40–50 cm. For plant in pot (see Fig. 5), the spacing is about 15–16 cm during early growth stage (plant width < 15 cm), and changed to 25–30 cm during next stage (plant width < 25 cm), for simultaneously considering 'High throughput phenotyping' and 'avoiding severe occlusion'.

5. Conclusion

This study achieved a low-cost and precise tomato seedling 3D phenotyping method. Tomato plant height and crown width calculation were successfully realized by sequentially performing steps, including 2D RGB image segmentation, 3D point cloud segmentation, and key points extraction. For plant segmentation in 2D images, the YOLO11-segment model achieved a mAP₅₀ of 96.0%. During this procedure, the vision language model GD could automatically assign each seedling a bounding box by using a text prompt 'plant'. This box information could also serve as the prompt of the SAM to generate the masks for tomato plant instance segmentation, which was further used to re-train a YOLO model. Therefore, very little annotation work (only some manual modification on a part of plant instances of GD-generated boxes) was required. The instance segmentation task was completed via weak labels input (manual input only involved a text prompt 'plant' and some boxes). Furthermore, with the 2D–3D coordinate mapping approach, 3D points of the plant instance could be easily segmented using the above-mentioned 2D masks from the YOLO model, achieving an Average Precision of 0.949. The presented new idea significantly decreased the complexity of 3D plant segmentation and the cost. It should also be highlighted that, for evaluation of plant with a sparse canopy, although an incomplete single-view point cloud of plants was used and simplified AI algorithms were utilized, the proposed tomato phenotyping method could still reach high performances on plant height (RMSE = 1.7 cm, R² = 0.93) and crown measurement (RMSE = 1.0 cm, R² = 0.95). For evaluation of tomato at different stage and other plant type with a dense canopy, the usage of a reference chessboard could further improve the height calculation. The developed method reduces the complexity of the data processing, and facilitates edge computing deployment, technology migration and application. It could be further applied to assist screening high-yield, stress-tolerant elite plants or varieties, which is crucial to intelligent tomato cultivation.

CRediT authorship contribution statement

Lei Zhou: Writing – original draft, Software, Methodology, Funding acquisition, Conceptualization. **Yingjun Xu:** Writing – review & editing, Visualization, Software, Data curation. **Chu Zhang:** Writing – review & editing, Investigation, Conceptualization. **Xiya Zhang:** Writing – review & editing, Resources. **Qian Wu:** Writing – review & editing, Resources. **Liming Bian:** Writing – review & editing, Visualization. **Osama Elsherbiny:** Writing – review & editing. **Huichun Zhang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Ao, Z.R., Wu, F.F., Hu, S.H., Sun, Y., Guo, Y.J., Guo, Q.H., Xin, Q.C., 2022. Automatic segmentation of stem and leaf components and individual maize plants in field terrestrial LiDAR data using convolutional neural networks. *Crop J.* 10 (5), 1239–1250. doi:10.1016/j.cj.2021.10.010.
- Bhattarai, U., Karkee, M., 2022. A weakly-supervised approach for flower/fruit counting in apple orchards. *Comput. Ind.* 138. doi:10.1016/j.compind.2022.103635 Article 103635.
- Bian, L.M., Zhang, H.C., Ge, Y.F., Cepl, J., Stejskal, J., El-Kassaby, Y.A., 2022. Closing the gap between phenotyping and genotyping: review of advanced, image-based phenotyping technologies in forestry. *Ann. For. Sci.* 79 (1). doi:10.1186/s13595-022-01143-x Article 22.
- Boogaard, F.P., van Henten, E.J., Kootstra, G., 2023. The added value of 3D point clouds for digital plant phenotyping—a case study on internode length measurements in cucumber. *Biosyst. Eng.* 234, 1–12. doi:10.1016/j.biosystemseng.2023.08.010.
- Cheng, Y.W., Ren, N., Hu, A.Q., Zhou, L.L., Qi, C., Zhang, S., Wu, Q., 2024. An improved 2D pose estimation algorithm for extracting phenotypic parameters of tomato plants in complex backgrounds. *Remote Sens.* 16 (23). doi:10.3390/rs16234385 Article 4385.
- Das, A., Pathan, F., Jim, J.R., Kabir, M.M., Mridha, M.F., 2025. Deep learning-based classification, detection, and segmentation of tomato leaf diseases: a state-of-the-art review. *Artif. Intell. Agric.* 15 (2), 192–220. doi:10.1016/j.iaia.2025.02.006.
- Deng, Y.B., Xi, H.R., Zhou, G.X., Chen, A.B., Wang, Y.F., Li, L.J., Hu, Y.H., 2023. An effective image-based tomato leaf disease segmentation method using MC-Unet. *Plant Phenom.* 5. doi:10.34133/plantphenomics.0049 Article 0049.
- Dong, S.F., Fan, X.Y., Li, X.H., Liang, Y.M., Zhang, M.Q., Yao, W., Yang, X.P., Wang, Z.P., 2025. Automatic 3D plant organ instance segmentation method based on PointNeXt and Quickshift plus. *Plant Phenom.* 7 (3). doi:10.1016/j.plaphe.2025.100065 Article 100065.
- Du, H., Yan, X.J., Wang, J.J., Xie, D., Pu, S.L., 2023. Rethinking the approximation error in 3D surface fitting for point cloud normal estimation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9486–9495. doi:10.1109/cvpr52729.2023.00915.
- Giménez-Gallego, J., Martínez-del-Rincon, J., González-Teruel, J.D., Navarro-Hellín, H., Navarro, P.J., Torres-Sánchez, R., 2024. On-tree fruit image segmentation comparing mask R-CNN and vision transformer models. Application in a novel algorithm for pixel-based fruit size estimation. *Comput. Electron. Agric.* 222. doi:10.1016/j.compag.2024.109077 Article 109077.
- Gong, L., Du, X.F., Zhu, K., Lin, K., Lou, Q.J., Yuan, Z., Huang, G.Q., Liu, C.L., 2021. Panicle-3d: efficient phenotyping tool for precise semantic segmentation of rice panicle point cloud. *Plant Phenom.* 2021. doi:10.34133/2021/9838929 Article 9838929.
- Han, X.W., Wang, H., Yuan, T., Zou, K.L., Liao, Q.F., Deng, K., Zhang, Z.Q., Zhang, C.L., Li, W., 2023. A rapid segmentation method for weed based on CDM and ExG index. *Crop Prot.* 172. doi:10.1016/j.cropro.2023.106321 Article 106321.
- Hao, J., Yan, Y., Zhang, Y., Zhang, Y.Y., Cao, Y.E., Wu, L.G., 2025. A cross-scale transfer learning framework: prediction of SOD activity from leaf microstructure to macroscopic hyperspectral imaging. *Plant Biotechnol. J.* 23 (4), 1091–1100. doi:10.1111/pbi.14566.
- Jin, S.C., Su, Y.J., Gao, S., Wu, F.F., Ma, Q., Xu, K.X., Hu, T.Y., Liu, J., Pang, S.X., Guan, H.C., Zhang, J., Guo, Q.H., 2020. Separating the structural components of maize for field phenotyping using terrestrial LiDAR data and deep convolutional neural networks. *IEEE Trans. Geosci. Remote Sens.* 58 (4), 2644–2658. doi:10.1109/tgrs.2019.2953092.
- Kanojia, A., Lyall, R., Sujeeth, N., Alseekh, S., Martínez-Rivas, F.J., Fernie, A.R., Gechev, T.S., Petrov, V., 2024. Physiological and molecular insights into the effect of a seaweed biostimulant on enhancing fruit yield and drought tolerance in tomato. *Plant Stress* 14. doi:10.1016/j.stress.2024.100692 Article 100692.
- Kasimatis, C.N., Psomakelis, E., Katsenios, N., Papatheodorou, M., Apostolou, D., Efthimiadou, A., 2025. Industrial tomato yield prediction using machine learning models. *Smart Agric. Technol.* 11. doi:10.1016/j.atech.2025.100920 Article 100920.
- Khanam, R., Hussain, M., 2024. YOLOv11: an overview of the key architectural enhancements. *arXiv preprint*. doi:10.48550/ARXIV.2410.17725.
- Kirillov, A., Mintun, E., Ravi, N., Mao, H.Z., Rolland, C., Gustafson, L., Xiao, T.T., Whitehead, S., Berg, A.C., Lo, W.Y., Dollár, P., Girshick, R., 2023. Segment anything. *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 3992–4003. doi:10.1109/iccv51070.2023.00371.
- Li, J.Y., Zhao, F.T., Zhao, H.M., Zhou, G.X., Xu, J.X., Gao, M.Z., Li, X., Dai, W.S., Zhou, H.L., Hu, Y.H., He, M.F., 2024. A multi-modal open object detection model for tomato leaf diseases with strong generalization performance using PDC-VLD. *Plant Phenom.* 6. doi:10.34133/plantphenomics.0220 Article 0220.
- Li, S.Z., Yan, Z.C., Ma, B.X., Guo, S.R., Song, H.X., 2025. Semantic segmentation method for high-resolution tomato seedling point clouds based on sparse convolution. *Agriculture-Basel* 15 (1). doi:10.3390/agriculture15010074 Article 74.
- Liang, X.M., Yu, W.B., Qin, L., Wang, J.F., Jia, P., Liu, Q., Lei, X.Y., Yang, M.L., 2025. Stem and leaf segmentation and phenotypic parameter extraction of tomato seedlings based on 3D point. *Agronomy Basel* 15 (1). doi:10.3390/agronomy15010120 Article 120.
- Liu, S.L., Zeng, Z.Y., Ren, T.H., Li, F., Zhang, H., Yang, J., Jiang, Q., Li, C.Y., Yang, J.W., Su, H., Zhu, J., Zhang, L., 2024. Grounding DINO: marrying DINO with grounded pre-training for open-set object detection. *18th European Conference on Computer Vision (ECCV)*, pp. 38–55. doi:10.1007/978-3-031-72970-6_3.
- Luo, L.Y., Jiang, X.T., Yang, Y., Samy, E.R.A., Lefsrud, M., Hoyos-Villegas, V., Sun, S.P., 2023. Eff-3DPeg: 3D organ-level plant shoot segmentation using annotation-efficient deep learning. *Plant Phenom.* 5. doi:10.34133/plantphenomics.0080 Article 0080.
- Ma, G.F., Javidan, S.M., Ampatzidis, Y., Zhang, Z., 2025. A novel hybrid technique for detecting and classifying hyperspectral images of tomato fungal diseases based on deep feature extraction and Manhattan distance. *Sensors* 25 (14). doi:10.3390/s25144285 Article 4285.

- Matsuura, Y., Zhang, H.M., Nakao, K., Qiong, C., Firmansyah, I., Kawai, S., Yamaguchi, Y., Maruyama, T., Hayashi, H., Nobuhara, H., 2023. High-precision plant height measurement by drone with RTK-GNSS and single camera for real-time processing. *Sci. Rep.* 13 (1). doi:[10.1038/s41598-023-32167-6](https://doi.org/10.1038/s41598-023-32167-6) Article 6329.
- Poudyal, D., Rosenqvist, E., Ottosen, C.O., 2019. Phenotyping from lab to field - tomato lines screened for heat stress using Fv/Fm maintain high fruit yield during thermal stress in the field. *Funct. Plant Biol.* 46 (1), 44–55. doi:[10.1071/fp17317](https://doi.org/10.1071/fp17317).
- Sangeetha, B., Krishnamoorthy, A.S., Renukadevi, P., Malathi, V.G., Sharmila, D.J.S., Amirtham, D., 2020. Antiviral activity of basidiomycetous fungi against groundnut bud necrosis virus in tomato. *Pestic. Biochem. Physiol.* 166. doi:[10.1016/j.pestbp.2020.104570](https://doi.org/10.1016/j.pestbp.2020.104570) Article 104570.
- Schönberger, J.L., Frahm, J.M., 2016. Structure-from-motion revisited. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4104–4113. doi:[10.1109/cvpr.2016.445](https://doi.org/10.1109/cvpr.2016.445).
- Shi, W.N., van de Zedde, R., Jiang, H.Y., Kootstra, G., 2019. Plant-part segmentation using deep learning and multi-view vision. *Biosyst. Eng.* 187, 81–95. doi:[10.1016/j.biosystemseng.2019.08.014](https://doi.org/10.1016/j.biosystemseng.2019.08.014).
- Shmaryahu, D., Lehman, R.L., Peleg, E., Shani, G., 2024. Estimating TYLCV resistance level using RGBD sensors in production greenhouse conditions. *Artif. Intell. Agric.* 14, 31–42. doi:[10.1016/j.iaia.2024.10.004](https://doi.org/10.1016/j.iaia.2024.10.004).
- Su, B.H., Jiang, B.Y., Yan, X.Y., Wu, T.R., He, J.Y., Yan, Z.Y., Qi, Y., Zhang, H., Xu, X.Y., Zhao, T.T., 2025. Analysis of the effect of improving drought tolerance via *SlGATA17* overexpression on tomato fruit quality. *Sci. Hortic.* 343. doi:[10.1016/j.scienta.2025.114083](https://doi.org/10.1016/j.scienta.2025.114083) Article 114083.
- Vayssade, J.A., Jones, G., Gée, C., Paoli, J.N., 2022. Pixelwise instance segmentation of leaves in dense foliage. *Comput. Electron. Agric.* 195. doi:[10.1016/j.compag.2022.106797](https://doi.org/10.1016/j.compag.2022.106797) Article 106797.
- Wang, Y.L., Zhang, Z.Y., 2025. Segment any leaf 3D: a zero-shot 3D leaf instance segmentation method based on multi-view images. *Sensors* 25 (2). doi:[10.3390/s25020526](https://doi.org/10.3390/s25020526) article 526.
- Wang, A.C., Qian, W.H., Li, A., Xu, Y.Z., Hu, J., Xie, Y.W., Zhang, L.Y., 2024a. NVW-YOLOv8s: an improved YOLOv8s network for real-time detection and segmentation of tomato fruits at different ripeness stages. *Comput. Electron. Agric.* 219. doi:[10.1016/j.compag.2024.108833](https://doi.org/10.1016/j.compag.2024.108833) Article 108833.
- Wang, L., Zhang, H.C., Bian, L.M., Zhou, L., Wang, S.Y., Ge, Y.F., 2024b. Poplar seedling varieties and drought stress classification based on multi-source, time-series data and deep learning. *Ind. Crop. Prod.* 218. doi:[10.1016/j.indcrop.2024.118905](https://doi.org/10.1016/j.indcrop.2024.118905) Article 118905.
- Ward, D., Moghadam, P., 2020. Scalable learning for bridging the species gap in image-based plant phenotyping. *Comput. Vis. Image Underst.* 197. doi:[10.1016/j.cviu.2020.103009](https://doi.org/10.1016/j.cviu.2020.103009) Article 103009.
- Wu, S.X., Su, H., Gao, F.Y., Yao, H.Y., Fan, X.L., Zhao, X.L., Li, Y.Y., 2023. An insight into the prevention and control methods for bacterial wilt disease in tomato plants. *Agronomy Basel* 13 (12). doi:[10.3390/agronomy13123025](https://doi.org/10.3390/agronomy13123025) Article 3025.
- Xie, K., Cui, C.X., Jiang, X., Zhu, J.Z., Liu, J.B., Du, A.B., Yang, W.N., Song, P., Zhai, R.F., 2025. Automated 3D segmentation of plant organs via the plant-MAE: a self-supervised learning framework. *Plant Phenom.* 7 (2). doi:[10.1016/j.plaph.2025.100049](https://doi.org/10.1016/j.plaph.2025.100049) Article 100049.
- Xu, B.H., Zhang, J.F., Tang, Z.X., Zhang, Y.S., Xu, L.L., Lu, H., Han, Z.G., Hu, W.J., 2025. Night-time environment enables robust field-based high-throughput plant phenotyping: a system platform and a case study on rice. *Comput. Electron. Agric.* 235. doi:[10.1016/j.compag.2025.110337](https://doi.org/10.1016/j.compag.2025.110337) Article 110337.
- Yadav, P.K., Bhujel, P., Bhandari, N., Sharma, S., Sharma, A., 2025. Screening tomato genotypes for early-stage drought tolerance using polyethylene glycol-induced osmotic stress. *BMC Plant Biol.* 25 (1). doi:[10.1186/s12870-025-07508-4](https://doi.org/10.1186/s12870-025-07508-4) article 1476.
- Yan, B., Li, X.M., 2024. RGB-D camera and fractal-geometry-based maximum diameter estimation method of apples for robot intelligent selective graded harvesting. *Fract. Fract.* 8 (11). doi:[10.3390/fractalfract8110649](https://doi.org/10.3390/fractalfract8110649) Article 649.
- Yang, N.Y., Huang, Z.Y., He, Y., Xiao, W.F., Yu, H., Qian, L.H., Xu, Y.X., Tao, Y.M., Lyu, P., Lyu, X.H., Feng, X.P., 2024. Detection of color phenotype in strawberry germplasm resources based on field robot and semantic segmentation. *Comput. Electron. Agric.* 226. doi:[10.1016/j.compag.2024.109464](https://doi.org/10.1016/j.compag.2024.109464) Article 109464.
- Yue, X., Qi, K., Na, X.Y., Zhang, Y., Liu, Y.H., Liu, C.H., 2023. Improved YOLOv8-seg network for instance segmentation of healthy and diseased tomato plants in the growth stage. *Agriculture-Basel* 13 (8). doi:[10.3390/agriculture13081643](https://doi.org/10.3390/agriculture13081643) article 1643.
- Zhang, H.C., Wang, L., Jin, X.L., Bian, L.M., Ge, Y.F., 2023a. High-throughput phenotyping of plant leaf morphological, physiological, and biochemical traits on multiple scales using optical sensing. *Crop J.* 11 (5), 1303–1318. doi:[10.1016/j.cj.2023.04.014](https://doi.org/10.1016/j.cj.2023.04.014).
- Zhang, Y., Xie, Y., Xu, X., Miao, M., 2023b. Measuring the cucumber leaf area using improved mask R-CNN. *Trans. Chin. Soc. Agric. Eng.* 39 (17), 182–189. doi:[10.11975/j.issn.1002-6819.202303149](https://doi.org/10.11975/j.issn.1002-6819.202303149).
- Zhang, Y.L., Xie, Y.L., Zhou, J.L., Xu, X.Y., Miao, M.M., 2024. Cucumber seedling segmentation network based on a multiview geometric graph encoder from 3D point clouds. *Plant Phenom.* 6. doi:[10.34133/plantphenomics.0254](https://doi.org/10.34133/plantphenomics.0254) Article 0254.
- Zhou, L., Zhang, H.C., Bian, L.M., Tian, Y., Zhou, H.P., 2024. Phenotyping of drought-stressed poplar saplings using exemplar-based data generation and leaf-level structural analysis. *Plant Phenom.* 6. doi:[10.34133/plantphenomics.0205](https://doi.org/10.34133/plantphenomics.0205) Article 0205.